Oppg 1

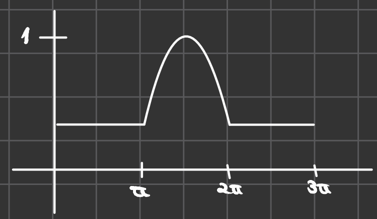
$$y''(t) + y(t) = \delta(t-\pi) + \delta(t-2\pi) \quad , \quad y(0) = y'(0) = 0$$

$$s^2 F(s) + sF(0) + F'(0) + F(s) = e^{-2\pi s} + e^{-\pi s}$$

$$F(s) = \frac{e^{-2\pi s} + e^{-\pi s}}{s^2+1} = \frac{e^{-2\pi s}}{s^2+1} + \frac{e^{-\pi s}}{s^2+1}$$

$$f(t) = \underbrace{\sin(t-2\pi)}_{=\sin(t)} u(t-2\pi) + \underbrace{\sin(t-\pi)}_{=-\sin(t)} u(t-\pi)$$

$$= (u(t-2\pi) - u(t-\pi))\sin(t)$$



Oppg 2

a)

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \Rightarrow \int f(x)\,dx = \sum_{n=-\infty}^{\infty} c_n \int_{-\pi}^{\pi} e^{inx}\,dx$$

$$f(x)e^{-inx} = \sum_{n=-\infty}^{\infty} c_n e^{inx} e^{-inx}$$

$$\int_{-\pi}^{\pi} f(x) e^{-inx}\,dx = \int_{-\pi}^{\pi} \sum_{n=-\infty}^{\infty} c_n\,dx$$

$$\int f(x)\cdot e^{-kxi} = \sum_{n=\infty} c_n \int_{-\pi}^{\pi} e^{inx}\cdot e^{-kxi}\,dx$$

$$\int f(x)\cdot e^{-nxi} = c_n \cdot 2\pi$$

$$\Rightarrow \frac{1}{2\pi}\int f(x)\cdot e^{-inx} = c_n$$

b)

$$f(x) = e^{-|x|} \quad , \quad x \in [-\pi, \pi]$$

$$c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-inx}\,dx$$

$$c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-|x|} e^{-inx}\,dx$$

$$= \frac{1}{2\pi}\left(\int_{-\pi}^{0} e^{x} e^{-inx}\,dx + \int_{0}^{\pi} e^{-x} e^{-inx}\,dx\right)$$

$$= \frac{1}{2\pi}\left(\int_{-\pi}^{0} e^{x(1-in)}\,dx + \int_{0}^{\pi} e^{-x(1+in)}\,dx\right)$$

$$= \frac{1}{2\pi}\left(\left[\frac{e^{x(1-in)}}{1-in}\right]_{-\pi}^{0} + \left[-\frac{e^{-x(1+in)}}{1+in}\right]_{0}^{\pi}\right)$$

$$= \frac{1}{2\pi}\left(\left(\frac{1}{1-in} - \frac{e^{-\pi}e^{+\pi in}}{1-in}\right) + \left(-\frac{e^{-\pi}e^{-\pi in}}{1+in} + \frac{1}{1+in}\right)\right)$$

$$= \frac{1}{2\pi}\left(\frac{1}{1-in} + \frac{1}{1+in} - \frac{e^{-\pi}(-1)^n}{1-in} - \frac{e^{-\pi}(-1)^n}{1+in}\right)$$

$$= \frac{1}{2\pi}\left(\frac{1+in}{1-n^2} + \frac{1-in}{1-n^2} - e^{-\pi}(-1)^n\left(\frac{1}{1-in} + \frac{1}{1+in}\right)\right)$$

$$= \frac{1}{2\pi}\left(\frac{2}{1-n^2} - e^{-\pi}(-1)^n\frac{2}{1-n^2}\right)$$

$$= \frac{1-e^{-\pi}(-1)^n}{\pi(1-n^2)}$$

Prob 3

$$u(x) = \begin{matrix} 0, & x<0 \\ 1, & x \geq 0 \end{matrix}$$

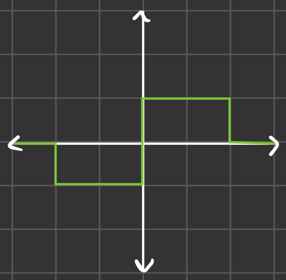
$$-u(x+1) + 2u(x) - u(x-1)$$

$x=-2: \Rightarrow 0$

$x=-1: \Rightarrow -1$

$x=0: \Rightarrow 1$

$x=1: \Rightarrow 0$



$$f(x) : \begin{cases} -1, & -1 \leq x < 0 \\ 1, & 0 \leq x < 1 \end{cases}$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, e^{-i\omega x} dx$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \left(\int_0^1 e^{-i\omega x} dx - \int_{-1}^0 e^{-i\omega x} dx \right)$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \left(\left[-\frac{e^{-i\omega x}}{i\omega} \right]_0^1 - \left[-\frac{e^{-i\omega x}}{i\omega} \right]_{-1}^0 \right)$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \left(-\frac{e^{-i\omega}}{i\omega} + \frac{1}{i\omega} - \left(-\frac{1}{i\omega} + \frac{e^{i\omega}}{i\omega} \right) \right)$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \left(\frac{2}{i\omega} - \frac{e^{-i\omega}}{i\omega} \right)$$

$$= \sqrt{\frac{2}{\pi}} \frac{1}{i\omega}$$

Prob 4

$$u_t = u_{xx} \qquad 0 \leq x \leq \pi, \quad t>0$$

$$u_x(0,t) = u_x(\pi,t) = 0, \quad u(x,0) = e^{-x}$$

$$u(x,t) = F(x)G(t)$$

$$F(x)G'(t) = F''(x)G(t)$$

$$\frac{F''(x)}{F(x)} = \frac{G'(t)}{G(t)} = k$$

$$F''(x) - kF(x) = 0, \quad k = -p^2, \quad p = \frac{n\pi x}{L} = 2nx$$

$$F''(x) + p^2 F(x) = 0$$

$$F(x) = A\cos(px) + B\sin(px) \Rightarrow F'(x) = -Ap\sin(px) + Bp\cos(px)$$

$$F(0) = Bp = 0 \Rightarrow B = 0$$

$$F(\pi) = -Ap\sin(p\pi) = 0 \ , \quad A \neq 0$$

$$O(n^2) = O(10^{-9}/2n)$$

$$F_n(x) = A\cos(\tfrac{n\pi x}{L})$$

$$2n^3 = 10^{-9}$$

$$G'(t) + p^2 G(t) = 0 \implies G(t) = G_0 e^{-\lambda^2 t} = G_0 e^{-4^2\pi^2 t}$$

0,0008 $\quad n = O(\varepsilon/2n)$

$$u(x,t) = \sum_{n=0}^{\infty} A_n \cos(2\pi x) e^{-4^2\pi^2 t}$$

0,00002

$$A_0 = \frac{1}{L}\int_0^L f(x)\,dx = \frac{2}{\pi}\int_0^{\frac{\pi}{2}} e^{-x}\,dx = -\frac{2}{\pi}\left[e^{-x}\right]_0^{\frac{\pi}{2}} = -\frac{2}{\pi}\left(1 - e^{-\frac{\pi}{2}}\right)$$

$$A_n = \frac{2}{L}\int_0^L f(x)\cos(\tfrac{n\pi x}{L})\,dx = \frac{4}{\pi}\int_0^{\pi/2} e^{-x}\cos(2nx)\,dx = \frac{2}{\pi}\int_0^{\frac{\pi}{2}} e^{-x}\left(e^{2ix} + e^{-2ix}\right)dx = \frac{2}{\pi}\int_0^{\frac{\pi}{2}} e^{x(2i-1)} + e^{-x(2i+1)}\,dx$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} e^{x(2i-1)} + e^{-x(2i+1)}\,dx = \frac{2}{\pi}\left[\frac{e^{x(2i-1)}}{2i-1} - \frac{e^{-x(2i-1)}}{2i+1}\right]_0^{\pi/2} = \frac{2}{\pi}\left(\frac{e^{\pi i - \frac{1}{2}}}{2i-1} - \frac{e^{-\pi i + \frac{1}{2}}}{2i+1} - \left(\frac{1}{2i-1} - \frac{1}{2i+1}\right)\right)$$